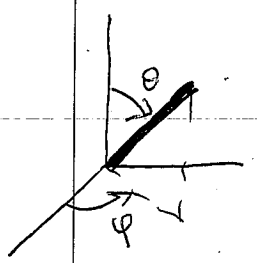
Magnetism is <u>absent</u> classically $\Rightarrow$ purely quantum effect.

Hydrogen atom

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\left(\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$



$$\psi(r,\theta,\varphi) = R(r)\,Y(\theta,\varphi)$$

<u>Angular equation:</u> $\sin\theta\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{\partial^2 Y}{\partial\varphi^2} = -l(l+1)\sin^2\theta\, Y$

spherical harmonics $\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$

$$Y_l^m(\theta,\varphi) = (\varepsilon)\sqrt{\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}}\; e^{im\varphi}\, P_l^m(\cos\theta)$$

(associated Legendre function)

$\varepsilon = 1$ for $m \le 0$
$\varepsilon = (-1)^m$ $m > 0$

$$Y_0^0 = \frac{1}{\sqrt{4\pi}} \qquad \int_0^{2\pi}\int_0^{\pi} |Y_l^m(\theta,\varphi)|^2 \sin\theta\, d\theta\, d\varphi = 1$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta = \sqrt{\frac{3}{4\pi}}\frac{z}{r}$$

$$Y_1^{\pm1} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\, e^{\pm i\varphi} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\frac{x\pm iy}{r}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$$

$$Y_2^{\pm1} = \mp\left(\frac{15}{8\pi}\right)^{1/2}\sin\theta\cos\theta\, e^{\pm i\varphi}$$

$$Y_2^{\pm2} = \left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\, e^{\pm i2\varphi}$$

Radial equation $\quad R(r) = \frac{u(r)}{r}$

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \underbrace{\left[V(r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right]}_{\text{effective potential}} u = E u$$

Hydrogen atom $\quad V(r) = -\frac{Ze^2}{4\pi\varepsilon_0}\frac{1}{r}$

(does not depend on l) $\quad E_n = -\left[\frac{m}{2\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2\right]\frac{1}{n^2} = \frac{E_1}{n^2} \qquad n = 1, 2, 3\ldots$ principal quantum number

$E_n = -\frac{\hbar^2}{2ma^2}\cdot\frac{1}{n^2}$ particle in a box of size a

Bohr radius $\quad a = \frac{4\pi\varepsilon_0\hbar^2}{me^2} = 0.53\cdot 10^{-10}\ m \quad (0.5\,\text{Å})$

(remember $\rho = r/an$) $\quad a = \frac{\hbar}{mc\alpha} \qquad \alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \simeq \frac{1}{137}$ fine structure constant

$$R_{10} = 2a^{-3/2}e^{-r/a}$$

$$R_{20} = \frac{1}{\sqrt{2}}a^{-3/2}\left(1 - \frac{r}{2a}\right)e^{-r/2a}$$

$$R_{21} = \frac{1}{\sqrt{24}}a^{-3/2}\left(\frac{r}{a}\right)e^{-r/2a}$$

p.154 $\quad R_{30} = \frac{2}{\sqrt{27}}a^{-3/2}\left(1 - \frac{2r}{3a} + \frac{2}{27}\frac{r^2}{a^2}\right)e^{-r/3a}$

etc.

Observe that $R_{nl} \sim e^{-\frac{r}{na}}$: $n>1$ levels are closer to the nuclei while $l>0$ states are "pushed" away from the center

Angular momentum of many-electron atom I

$$\hat{l}_z = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = x p_y - y p_x$$

$$\hat{L}_z = -i\hbar \sum_i \left(x_i \frac{\partial}{\partial y_i} - y_i \frac{\partial}{\partial x_i}\right)$$

Suppose 1-electron atom (Hydrogen)

$$H_1 = -\frac{\hbar^2}{2m}\vec{\nabla}^2 - \frac{e^2}{4\pi\varepsilon_0 r} = \frac{\vec{p}^2}{2m} - \frac{e^2}{4\pi\varepsilon_0 r}$$

$$\hat{l}_z \frac{1}{r} = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)\frac{1}{\sqrt{x^2+y^2+z^2}} =$$

$$= -i\hbar\left(x\frac{-\frac{1}{2}\cdot 2y}{(x^2+y^2)^{3/2}} - y\frac{-\frac{1}{2}\cdot 2x}{(x^2+y^2)^{3/2}}\right) = -i\hbar\frac{-xy+yx}{(x^2+y^2)^{3/2}} = 0$$

Obviously $\hat{l}_{x,y}\frac{1}{r} = 0$

In fact, for arbitrary function of $|\vec{r}| = r$.

$$\hat{\vec{l}} f(r) = 0$$

$$\vec{l} = \vec{r} \times \frac{\hbar}{i}\vec{\nabla} \quad \text{but} \quad \vec{\nabla} f(r) = \hat{u}_r \frac{d}{dr} f(r)$$

$$\Rightarrow \vec{l} f(r) = (-i\hbar)(\vec{r} \times \hat{u}_r)\frac{d}{dr} f(r) = 0$$

$\Rightarrow$ angular momentum is associated with non-trivial angular dependence of the (wave) function.

Thus, $\vec{l}$ and $f(r)$ commute:

$$[\vec{l}, f]\psi = (\vec{l} f(r) - f(r)\vec{l})\psi = \underbrace{\vec{l} f(r)\psi}_{\varepsilon_{ijk} x_j \hat{p}_k (f\psi) =} - f(r)\vec{l}\psi =$$

$$= \underbrace{(\vec{l} f)}_{0}\psi + f\vec{l}\psi - f\vec{l}\psi = 0$$

$$= \varepsilon_{ijk}(x_j \hat{p}_k f)\psi + f(\varepsilon_{ijk} x_j \hat{p}_k \psi)$$

$$= f\vec{l}\psi$$

Check that $[\vec{l}_z, \vec{p}^2] = 0$

First, $[A, B^2] = AB^2 - B^2A = AB^2 - B^2A + BAB - BAB =$

$= (AB - BA)B + B(AB - BA) = [A,B]B + B[A,B]$

$$[\hat{l}_z, \vec{p}] = [\hat{l}_z, \hat{p}_x \hat{x} + \hat{p}_y \hat{y} + \hat{p}_z \hat{z}] =$$

$$= [x p_y - y p_x, p_x]\hat{x} + [x p_y - y p_x, p_y]\hat{y} =$$

$$= \underbrace{[x p_y, p_x]}_{p_y \underbrace{[x, p_x]}_{i\hbar}} \hat{x} + \underbrace{[y p_x, p_y]}_{p_x \underbrace{[y, p_y]}_{i\hbar}} \hat{y} = i\hbar(p_y \hat{x} - p_x \hat{y}) =$$

$$= i\hbar\, \varepsilon_{zij}\, p_j\, \hat{e}_i$$

$$\varepsilon_{zxy} = 1 \qquad \varepsilon_{zyx} = -1$$

More generally $[l_i, p_k] = i\hbar\, \varepsilon_{ikl}\, p_l$ $\Rightarrow$ definition of the vector operator!

$$[l_z, \vec{p}^2] = \underbrace{[l_z, p_x]}_{i\hbar\, \varepsilon_{zxy}\, p_y \cdot p_x} p_x + p_x \underbrace{[l_z, p_x]}_{i\hbar\, \varepsilon_{zxy}\, p_x \cdot p_y}$$

$$+ \underbrace{[l_z, p_y]}_{i\hbar\, \varepsilon_{zyx}\, p_x p_y} p_y + \underbrace{p_y [l_z, p_y]}_{i\hbar\, \varepsilon_{zyx}\, p_y p_x} + \underbrace{[l_z, p_z]}_{i\hbar\, \cancel{\varepsilon_{zzi}}_{=0}\, p_i p_z} p_z + p_z \underbrace{[l_z, p_z]}_{=0}$$

$$= 2i\hbar\, p_x p_y (\varepsilon_{zxy} + \varepsilon_{zyx}) = 0$$

Thus $[l_z, H_1] = 0$. Clearly $[l_{x,y}, H_1] = 0$

and $[\vec{l}^2, H_1] = [l_x^2 + l_y^2 + l_z^2, H_1] = 0$

Now 2-electron atom:

$$\vec{L} = \vec{l}_1 + \vec{l}_2 \quad , \quad L_z = l_{1z} + l_{2z}$$

$$H_2 = \underbrace{\frac{\vec{p}_1^{\,2}}{2m} - \frac{e^2}{4\pi\varepsilon_0 r_1}}_{H_1^{(1)}} + \underbrace{\frac{\vec{p}_2^{\,2}}{2m} - \frac{e^2}{4\pi\varepsilon_0 r_2}}_{H_1^{(2)}} + \underbrace{\frac{e^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_2|}}_{\text{inter-electron repulsion}}$$

Obviously $[l_{1z}, H_1^{(1)}] = 0$ and $[l_{1z}, H_1^{(2)}] = 0$ trivially.

Also $[l_{2z}, H_1^{(1)}] = 0$ $\quad [l_{2z}, H_1^{(2)}] = 0$.

How about $(l_{1z} + l_{2z}) \dfrac{1}{|\vec{r}_1 - \vec{r}_2|} = \left[-i\hbar\left(x_1 \dfrac{\partial}{\partial y_1} - y_1 \dfrac{\partial}{\partial x_1}\right) + \right.$

$$\left. + (-i\hbar)\left(x_2 \frac{\partial}{\partial y_2} - y_2 \frac{\partial}{\partial x_2}\right)\right] \frac{1}{\sqrt{(x_1-x_2)^2 + (y_1-y_2)^2 + (z_1-z_2)^2}} =$$

$$= (-i\hbar)\left\{ \frac{x_1 \left(-\frac{1}{2}\right) 2(y_1-y_2) - y_1 \left(-\frac{1}{2}\right) 2(x_1-x_2)}{\left[(x_1-x_2)^2 + (y_1-y_2)^2 + (z_1-z_2)^2\right]^{3/2}} \right.$$

$$\left. + \frac{x_2 \left(-\frac{1}{2}\right) 2(y_1-y_2)(-1) - y_2 \left(-\frac{1}{2}\right) 2(x_1-x_2)(-1)}{[\quad]^{3/2}} \right\}$$

$$= -i\hbar \; \frac{-x_1(y_1-y_2) + y_1(x_1-x_2) + x_2(y_1-y_2) - y_2(x_1-x_2)}{|\vec{r}_1 - \vec{r}_2|^3}$$

$$= \frac{-i\hbar}{|\vec{r}_1 - \vec{r}_2|^3}\left(x_1 y_2 - y_1 x_2 + x_2 y_1 - y_2 x_1\right) = 0$$

So that $(l_{1a} + l_{2a}) \dfrac{1}{|\vec{r}_1 - \vec{r}_2|} = 0$ for any $a = x, y, z$.

Thus <u>total</u> L_z is conserved:

$$[L_z, H_z] = 0 \quad \text{and} \quad [L_x, H_z] = [L_y, H_z] = 0.$$

Clearly $[\vec{L}^2, H_z] = 0$.

Generally speaking, follows from angular isotropy of space.

$\Rightarrow$ state of N-electron atom is characterized by energy, <u>total L_z</u> and <u>total $\vec{L}^2$</u>:

$$\hat{H} \, |n, \mathcal{L}, M\rangle = E_{n\mathcal{L}m} \, |n, \mathcal{L}, M\rangle$$

$$\hat{L}_z \, |n, \mathcal{L}, M\rangle = \hbar M \, |n, \mathcal{L}, M\rangle$$

$$\vec{L}^2 \, |n, \mathcal{L}, M\rangle = \hbar^2 \mathcal{L}(\mathcal{L}+1) \, |n, L, M\rangle$$

here $L_z = \sum_{i=1}^{N} l_{iz}$ and $M = \sum_{i=1}^{N} m_i$.